Middle East Technical University

Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 15: Nonlinear Resistive Circuits

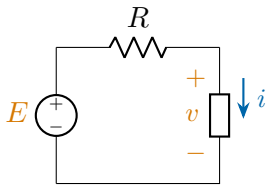
M. Mert Ankaralı

Colour code used in these notes and in the slides

current i voltage v element characteristic - - - load line

Motivation. A resistor is defined by its characteristic, a curve in the v–i plane. An LTI resistor is a straight line through the origin, and every method so far (node and mesh analysis, Thévenin equivalents, superposition) relies on that. Diodes, lamps and transistors have bent characteristics. This lecture solves circuits with **one nonlinear resistor** by the **load line**, uses the **ideal diode** to build **piecewise-linear** characteristics, and ends with a short look at **linear time-varying** circuits.

1. A circuit with one nonlinear resistor



A source E and a resistor R drive a nonlinear resistor with the characteristic $i = g(v)$ (a voltage-controlled resistor, see Section 3.2). KVL gives

$$E = R\,i + v.$$

The problem: find the pair (v_o, i_o), the **operating point**.

1.1. The algebraic solution

Substitute the characteristic into KVL and solve one equation in one unknown:

$$E = R\,g(v) + v \quad \Rightarrow \quad v_o, \qquad i_o = g(v_o).$$

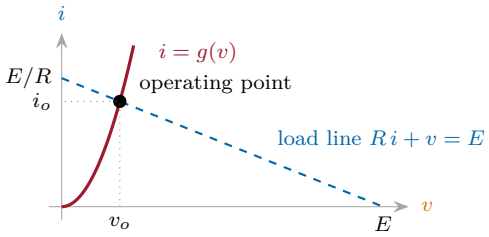
This needs a formula for g. Often the characteristic is only a measured curve.

1.2. The graphical solution: the load line

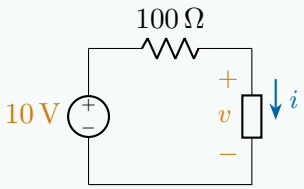
The pair (v, i) satisfies two constraints at the same time:

- the circuit: $R\,i + v = E$, a straight line, the **load line**;
- the element: $i = g(v)$.

The operating point is where the two meet. The load line crosses the axes at $v = E$ and $i = E/R$.



Example 15.1 — the operating point



The nonlinear resistor has the characteristic

$$i = \begin{cases} \frac{3}{100}\,v^2, & v \geq 0, \\ 0, & v < 0. \end{cases}$$

Find the operating point.

Solution

Step 1. KVL

$$100\,i + v - 10 = 0.$$

Step 2. The case $v < 0$

Here $i = 0$, so KVL gives $v = 10\,\text{V}$. This contradicts $v < 0$: there is no solution with $v < 0$.

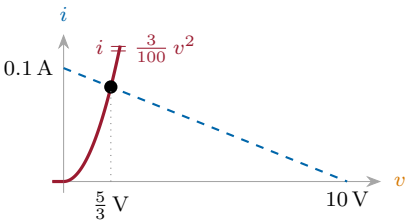
Step 3. The case $v \geq 0$

Plug $i = \frac{3}{100}\,v^2$ into KVL:

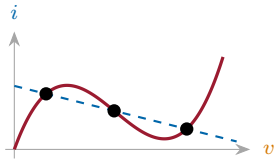
$$3\,v^2 + v - 10 = 0$$
$$(3\,v - 5)(v + 2) = 0.$$

Only $v = \frac{5}{3}\,\text{V}$ satisfies $v \geq 0$. Then $i = \frac{3}{100} \cdot \frac{25}{9} = \frac{1}{12}\,\text{A}$:

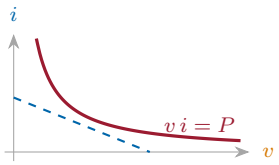
$$(v_o, i_o) = (\tfrac{5}{3}\,\text{V},\ \tfrac{1}{12}\,\text{A}).$$



1.3. None, one or many operating points



many



none

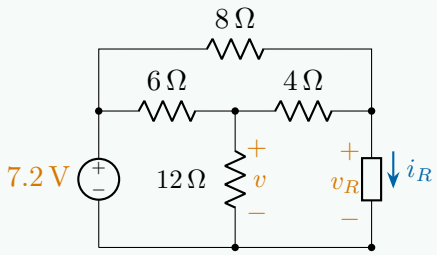
A linear circuit with a unique solution has exactly one. A nonlinear one need not: a bent curve can meet the load line several times, and a curve such as $v\,i = P$ may not meet it at all.

2. A nonlinear resistor in a larger circuit

When the rest of the circuit is linear, three steps reduce the problem to the one of Section 1:

1. find the **Thévenin equivalent** of the linear part, seen by the nonlinear resistor;
2. find the **operating point** (v_R, i_R) with the load line;
3. by the **substitution theorem**, replace the nonlinear resistor by a voltage source v_R (or a current source i_R); no other voltage or current in the circuit changes, and every other variable then follows from a linear circuit.

Example 15.2 — Thévenin, operating point, substitution



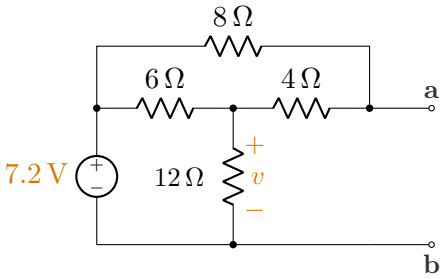
The nonlinear resistor has the characteristic

$$i_R = \begin{cases} \frac{1}{4}\,v_R{}^2, & v_R \geq 0, \\ 0, & v_R < 0. \end{cases}$$

Find v.

Solution

Step 1. The Thévenin equivalent



Remove the nonlinear resistor; its terminals become the port a–b. Killing the 7.2 V source (a short):

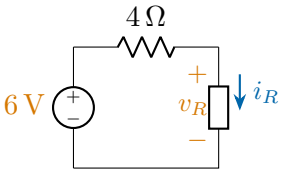
$$R_{\text{th}} = 8 \parallel (4 + 6 \parallel 12)$$
$$= 8 \parallel 8 = 4\,\Omega.$$

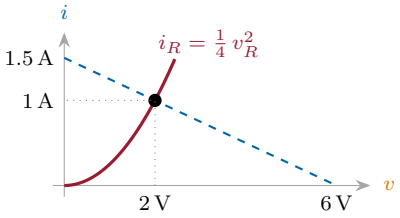
For v_{oc}, take b as the ground; with node voltages e_1 (the top of 12 Ω) and e_a:

$$\text{Node 1:} \quad \frac{e_1 - 7.2}{6} + \frac{e_1}{12} + \frac{e_1 - e_a}{4} = 0$$
$$\text{Node a:} \quad \frac{e_a - 7.2}{8} + \frac{e_a - e_1}{4} = 0$$

These give $e_1 = 5.4\,\text{V}$ and $v_{\text{th}} = v_{\text{oc}} = e_a = 6\,\text{V}$.

Step 2. The operating point



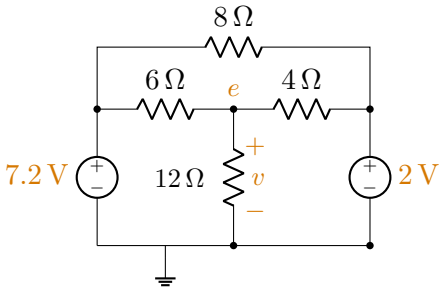


KVL: $4\,i_R + v_R - 6 = 0$. Plug $i_R = \frac{1}{4}\,{v_R}^2$:

$$ {v_R}^2 + v_R - 6 = 0 $$
$$ (v_R + 3)(v_R - 2) = 0. $$

With $v_R \geq 0$: $v_R = 2\,\text{V}$ and $i_R = 1\,\text{A}$.

Step 3. Substitution



Replace the nonlinear resistor by a 2 V source. One unknown node voltage e remains:

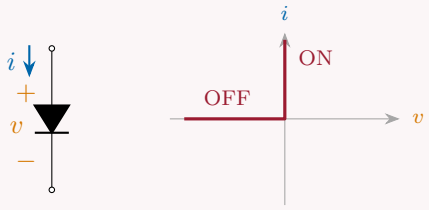
$$ \frac{e - 7.2}{6} + \frac{e}{12} + \frac{e - 2}{4} = 0. $$

Multiply by 12: $2\,e - 14.4 + e + 3\,e - 6 = 0$, so

$$ v = e = 3.4\,\text{V}. $$

3. The ideal diode

Ideal diode



ON: $v = 0$ and $i \geq 0$: a short circuit.

OFF: $i = 0$ and $v \leq 0$: an open circuit.

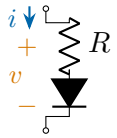
The characteristic is made of two straight pieces.

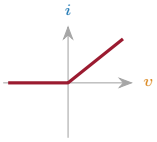
Turning an element around. If the two terminals of any resistive element are swapped, both v and i change sign: the characteristic is reflected through the origin. A reversed diode conducts for $i \leq 0$ and blocks for $v \geq 0$.

Series and parallel. In series the elements carry the same current, so the voltages add: the curves are added *horizontally*. In parallel the elements share the same voltage, so the currents add: the curves are added *vertically*. Combining ideal diodes, LTI resistors and constant sources this way always gives a **piecewise-linear** characteristic.

3.1. Simple series and parallel

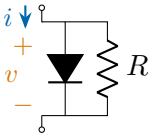
Start with one diode and one resistor R. In series, the diode curve is added horizontally to the line of R; in parallel, vertically.

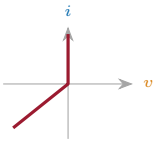




I: R and a diode in series

$i > 0$: $v = Ri$; $i = 0$: $v \leq 0$





II: R and a diode in parallel

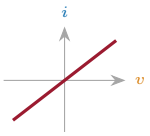
$v < 0$: $i = v/R$; $v = 0$: $i \geq 0$

3.2. Reminder: voltage-controlled or current-controlled

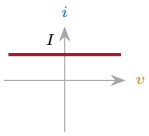
Recall: a characteristic is a curve in the v–i plane.

- **Voltage-controlled:** $i = g(v)$. Each *vertical* line meets the curve at most once.
- **Current-controlled:** $v = r(i)$. Each *horizontal* line meets the curve at most once.

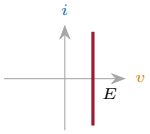
A curve can pass both tests, one of them, or neither:



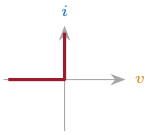
resistor
both



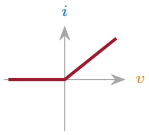
current source
voltage-controlled



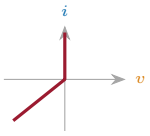
voltage source
current-controlled



ideal diode
neither



Case I: R in series
voltage-controlled



Case II: R in parallel
current-controlled

The ideal diode alone is neither. A resistor in series removes its vertical piece (Case I); a resistor in parallel removes its horizontal piece (Case II).

Why it matters.

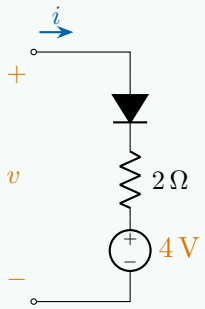
- **Which analysis.** If $i = g(v)$, the current goes straight into KCL: node analysis fits. If $v = r(i)$, the voltage goes straight into KVL: mesh analysis fits. Example 15.1 uses $i = g(v)$ for this reason.
- **Series and parallel.** In series, current-controlled elements give $v = r_1(i) + r_2(i)$. In parallel, voltage-controlled elements give $i = g_1(v) + g_2(v)$. A curve that is neither is added point by point, as in Cases I and II.

- **The ideal diode.** It is neither, so no single formula describes it. This is why we assume its state (ON or OFF) and check the result.
- **The operating point.** A curve that never goes down (a resistor, a diode, Cases I and II) meets a load line at most once. Several operating points need a curve that goes down somewhere, as in the figure with many operating points.

3.3. Assume the state of the diode

A diode is either ON (a short) or OFF (an open). With a state assumed, the circuit is linear. The **assumed-state method**: assume a state, solve the linear circuit, and keep the result only where the assumption holds ($i_D \geq 0$ for ON, $v_D < 0$ for OFF).

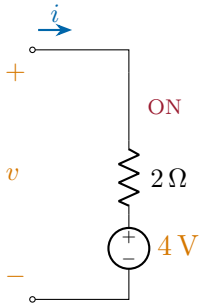
Example 15.3 — a diode, a resistor and a source



The one-port is a diode, a 2 Ω resistor and a 4 V source in series. Find its i–v characteristic, first by assuming states and then graphically.

Solution

Step 1. Assume the diode ON

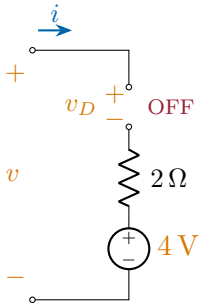


The diode is a short. KVL:

$$v = 2\,i + 4.$$

The diode current is $i_D = i = (v - 4)/2$. The condition $i_D \geq 0$ holds for $v \geq 4\,\text{V}$.

Step 2. Assume the diode OFF



The diode is an open, so $i = 0$ and the 2 Ω resistor has no voltage. KVL:

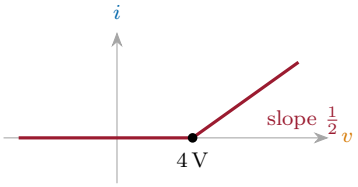
$$v_D = v - 4.$$

The condition $v_D < 0$ holds for $v < 4\,\text{V}$.

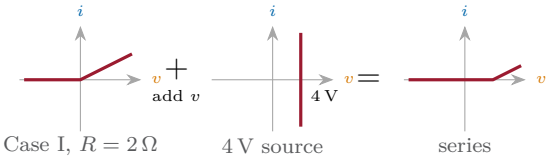
Step 3. The characteristic

$$i = \begin{cases} \dfrac{v-4}{2}, & v \geq 4\,\mathrm{V}, \\ 0, & v < 4\,\mathrm{V}. \end{cases}$$

The two ranges do not overlap and together cover every v. The breakpoint $(4\,\mathrm{V}, 0)$ is where $i_D = 0$ and $v_D = 0$ at the same time.



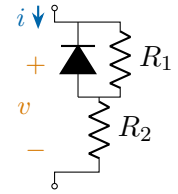
Step 4. Graphically

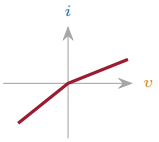


The diode and $2\,\Omega$ are Case I. The source is a vertical line at $v = 4\,\mathrm{V}$; adding it horizontally shifts the curve right by $4\,\mathrm{V}$: the same result. ✓

3.4. More connections

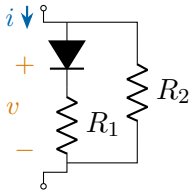
The same reasoning gives the characteristics of the connections below. In each one, find the piece with the diode ON, the piece with it OFF, and the breakpoint. For simplicity, Cases III and IV use two equal resistors: $R_1 = R_2 = R$.

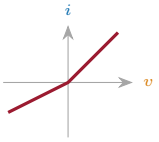




III: reversed diode $\|$ R_1, then R_2

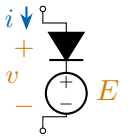
slope $\frac{1}{2R}$ for $i > 0$, $\frac{1}{R}$ for $i < 0$

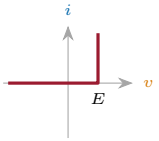




IV: diode and R_1, in parallel with R_2

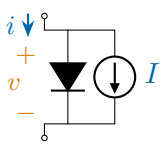
slope $\frac{2}{R}$ for $v > 0$, $\frac{1}{R}$ for $v < 0$

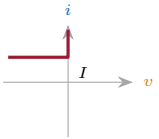




V: a diode and a source E in series

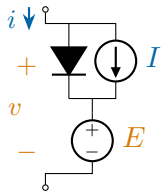
the diode curve shifted right by E

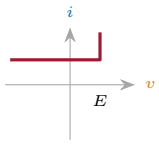




VI: a diode and a source I in parallel

the diode curve shifted up by I



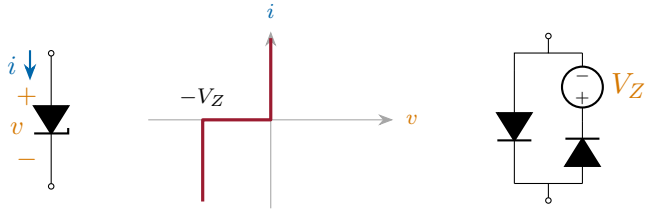


VII: Case VI in series with E (+ at the top)

Case VI shifted right by E

A voltage source in series shifts a curve horizontally; a current source in parallel shifts it vertically.

3.5. The ideal Zener diode



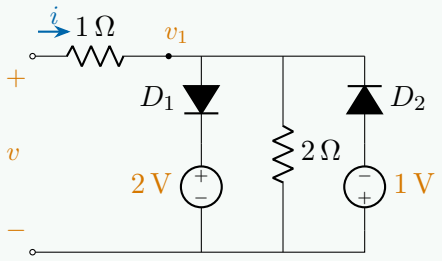
In the forward direction the Zener diode is an ideal diode. In the reverse direction it blocks until $v = -V_Z$, and then holds that voltage. It is equivalent to a diode in parallel with a reversed diode and a battery V_Z in series.

4. Piecewise-linear circuits

4.1. The characteristic of a circuit

With n diodes there are 2^n combinations of states, and each gives a linear circuit. Solve each one, keep it where its conditions hold, and drop the impossible ones. Each diode adds at most one breakpoint.

Example 15.4 — two diodes



A 1 Ω resistor is in series with three parallel branches: D_1 with a 2 V source, a 2 Ω resistor, and a reversed D_2 with a 1 V source. v_1 is the voltage of the parallel part. Find the i–v characteristic of the one-port.

Solution

Step 1. Both diodes OFF

Only the 1 Ω and 2 Ω resistors carry current, in series:

$$v_1 = 2\,i, \qquad v = 3\,i.$$

The diode voltages are $v_{D_1} = v_1 - 2$ and $v_{D_2} = -1 - v_1$. Both are negative for $-1 < v_1 < 2$, that is $-0.5 < i < 1$, or $-1.5\,\text{V} < v < 3\,\text{V}$.

Step 2. D_1 **ON,** D_2 **OFF**

D_1 is a short, so $v_1 = 2\,\text{V}$ and

$$v = i + 2.$$

The $2\,\Omega$ resistor carries 1 A, so $i_{D_1} = i - 1$. The condition $i_{D_1} \geq 0$ holds for $i \geq 1$, or $v \geq 3\,\text{V}$. Check D_2: $v_{D_2} = -1 - 2 < 0$. ✓

Step 3. D_2 **ON,** D_1 **OFF**

D_2 is a short, so $v_1 = -1\,\text{V}$ and

$$v = i - 1.$$

The $2\,\Omega$ resistor carries $-0.5\,\text{A}$. KCL at the top node gives the forward current of D_2 (upward): $i_{D_2} = -0.5 - i$. The condition $i_{D_2} \geq 0$ holds for $i \leq -0.5$, or $v \leq -1.5\,\text{V}$. Check D_1: $v_{D_1} = -1 - 2 < 0$. ✓

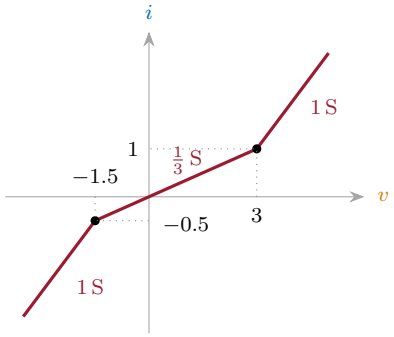
Step 4. Both diodes ON

This would need $v_1 = 2$ and $v_1 = -1$ at the same time: impossible.

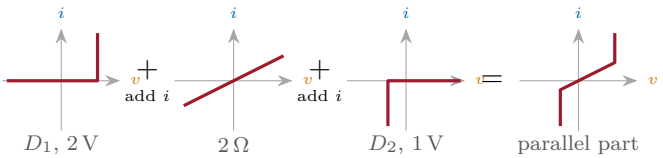
Step 5. The characteristic

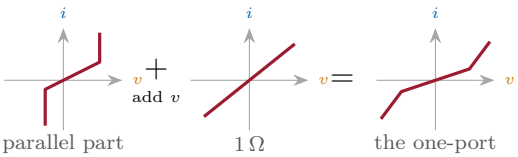
$$i = \begin{cases} v - 2, & v \geq 3\,\text{V}, \\ v/3, & -1.5\,\text{V} \leq v \leq 3\,\text{V}, \\ v + 1, & v \leq -1.5\,\text{V}. \end{cases}$$

Two diodes give two breakpoints: $(-1.5\,\text{V}, -0.5\,\text{A})$ and $(3\,\text{V}, 1\,\text{A})$.



Step 6. Graphically



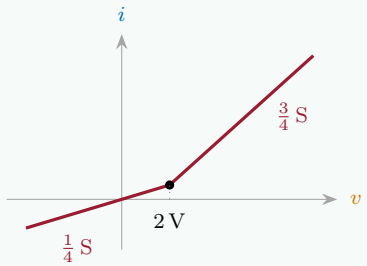


The first branch is Case V: vertical at $v_1 = 2\,\text{V}$. The third is Case V turned around: vertical at $v_1 = -1\,\text{V}$. Adding the three currents gives the parallel part: slope $\frac{1}{2}$ between the two vertical pieces. Adding the $1\,\Omega$ horizontally turns the slopes into $\frac{1}{1+2} = \frac{1}{3}$ in the middle and 1 outside. ✓

4.2. A circuit from a characteristic

The reverse problem: given a piecewise-linear characteristic, find a circuit. Split the curve into pieces whose circuits are known (the basic connections above), using horizontal addition (series) or vertical addition (parallel), and connect the pieces. Each breakpoint needs a diode. The answer is **not unique**: different circuits can have the same characteristic.

Example 15.5 — one breakpoint



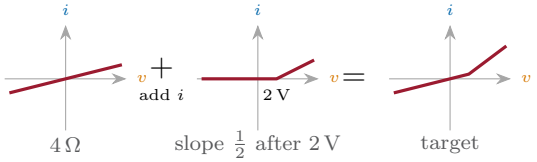
The characteristic is $i = v/4$ for $v \le 2\,\mathrm{V}$ and has slope $\frac{3}{4}\,\mathrm{S}$ for $v \ge 2\,\mathrm{V}$. Find a circuit with this characteristic.

Solution

Step 1. Count the breakpoints

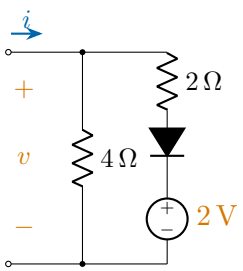
There is one breakpoint, at $(2\,\mathrm{V}, 0.5\,\mathrm{A})$, so one diode is enough. Below it the curve is a line through the origin with slope $\frac{1}{4}\,\mathrm{S}$: a $4\,\Omega$ resistor.

Step 2. Split it in parallel (add currents)



Keep the $4\,\Omega$ resistor for all v. Above $2\,\mathrm{V}$ the slope must grow by $\frac{3}{4} - \frac{1}{4} = \frac{1}{2}\,\mathrm{S}$. A branch with zero current below $2\,\mathrm{V}$ and slope $\frac{1}{2}\,\mathrm{S}$ above is Case I with $R = 2\,\Omega$, shifted right by a $2\,\mathrm{V}$ source in series. Since the currents add, this branch goes in parallel with the $4\,\Omega$ resistor.

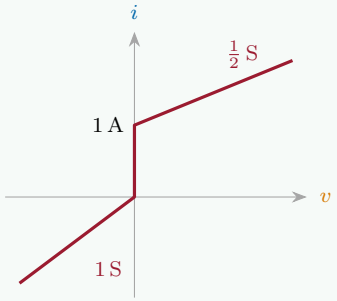
Step 3. The circuit



A $4\,\Omega$ resistor in parallel with a branch of $2\,\Omega$, a diode and a $2\,\mathrm{V}$ source. Check at $v = 4\,\mathrm{V}$: the resistor carries $1\,\mathrm{A}$ and the branch $(4-2)/2 = 1\,\mathrm{A}$, so $i = 2\,\mathrm{A}$; the target gives $0.5 + \frac{3}{4}\cdot 2 = 2\,\mathrm{A}$. ✓ The elements of the branch can be put in any order: the circuit is not unique.

The next example needs two diodes and both kinds of addition.

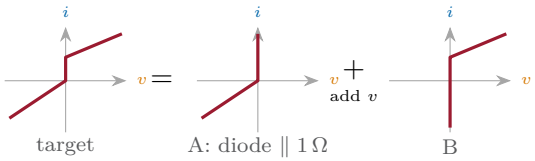
Example 15.6 — a circuit with two breakpoints



The characteristic has slope 1 S for $v < 0$, a vertical piece from 0 to 1 A at $v = 0$, and slope $\frac{1}{2}$ S above 1 A. Find a circuit with this characteristic.

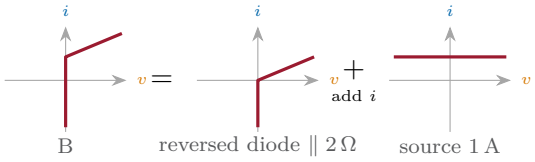
Solution

Step 1. Split it in series (add voltages)



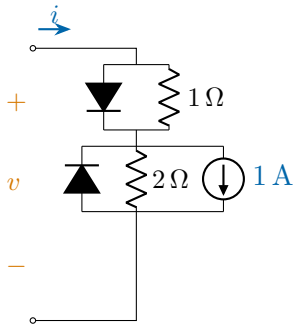
Piece A is Case II with $R = 1\,\Omega$: it gives the slope 1 S for $v < 0$ and is a short for $i \geq 0$. Piece B is a short for $i \leq 1$ A and has slope $\frac{1}{2}$ S above 1 A.

Step 2. Split B in parallel (add currents)



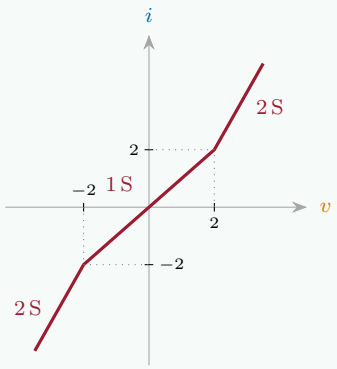
A reversed diode in parallel with 2 Ω is Case II reflected through the origin. A 1 A current source in parallel lifts it by 1 A (Case VI).

Step 3. The circuit



Piece A (a diode in parallel with 1 Ω) in series with piece B (a reversed diode, 2 Ω and the 1 A source, all in parallel). Check at $v = 2$ V: the top diode is ON (a short), the bottom diode is OFF, and $i = 1 + 2/2 = 2$ A. ✓

Example 15.7 — a numerical characteristic



The characteristic passes through the origin with slope 1 S for $-2\,\mathrm{V} < v < 2\,\mathrm{V}$ and slope 2 S outside this interval. Find a circuit.

Solution

Step 1. Split it in parallel (add currents)

All pieces share the voltage v, so write the current as a sum:

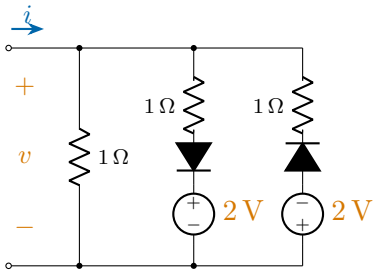
- a 1 Ω resistor: the slope 1 S in the middle;
- a branch with current $v - 2$ for $v > 2\,\mathrm{V}$ and zero otherwise;
- a branch with current $v + 2$ for $v < -2\,\mathrm{V}$ and zero otherwise.

Outside $\pm 2\,\mathrm{V}$ one branch adds 1 S, so the slope becomes 2 S.

Step 2. The two branches

The second piece is like the branch of Example 15.5, now with 1 Ω: a resistor, a diode and a 2 V source in series, conducting only when $v > 2\,\mathrm{V}$. The third piece is its reflection through the origin: the same branch turned around.

Step 3. The circuit

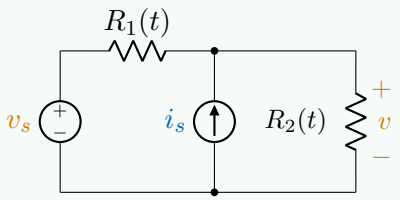


Check at $v = 3\,\mathrm{V}$: the resistor carries 3 A, the middle branch $3 - 2 = 1\,\mathrm{A}$ and the reversed branch 0; in total 4 A. The target gives $2 + 2 \cdot 1 = 4\,\mathrm{A}$. ✓

5. Linear time-varying circuits

A linear time-varying (LTV) resistor has the characteristic $v = R(t)\,i$: at each instant a line through the origin, but its slope changes with time. Such circuits are not nonlinear, and at each instant they are solved like LTI circuits.

Example 15.8 — a linear time-varying circuit



$R_1(t)$ and $R_2(t)$ are linear time-varying resistors. Express v in terms of the inputs v_s and i_s. Does superposition still hold?

Solution

Step 1. KCL at the top node

$$\frac{v - v_s}{R_1(t)} - i_s + \frac{v}{R_2(t)} = 0.$$

Step 2. Solve for v

$$v = \underbrace{\frac{R_2(t)}{R_1(t) + R_2(t)}}_{k_1(t)} v_s + \underbrace{\frac{R_1(t)\,R_2(t)}{R_1(t) + R_2(t)}}_{k_2(t)} i_s = \mathbf{k}(t)\,\mathbf{u},$$

with the input vector $\mathbf{u} = [\,v_s \;\; i_s\,]^\mathsf{T}$ and $\mathbf{k}(t) = [\,k_1(t) \;\; k_2(t)\,]$.

Step 3. Superposition

For input vectors $\mathbf{u}$, $\mathbf{w}$ and numbers α, β, $v(\alpha\,\mathbf{u}+\beta\,\mathbf{w}) = \alpha\,v(\mathbf{u})+\beta\,v(\mathbf{w})$: superposition still holds for LTV resistive circuits. Only the coefficients k_1, k_2 now depend on time.

Summary

In the circuit	What to do
one nonlinear resistor	solve $E = R\,g(v) + v$, or intersect the load line $R\,i + v = E$ with $i = g(v)$
a linear rest	Thévenin equivalent, operating point, then the substitution theorem
ideal diodes	assume ON ($v_D = 0$, $i_D \geq 0$) or OFF ($i_D = 0$, $v_D < 0$), solve, check
series, parallel connections	add the curves horizontally (voltages) or vertically (currents)
voltage- or current-controlled	vertical line test: $i = g(v)$, into KCL; horizontal line test: $v = r(i)$, into KVL
a given characteristic	split it into known pieces; each breakpoint needs a diode; not unique
linear time-varying resistors	$v = \mathbf{k}(t)\,\mathbf{u}$: superposition still holds